

Causes of variation in vaccine efficacy in a wild rodent population

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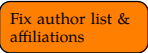
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TODO LIST

1	 Fix author list & affiliations	1	
2	Individuals vary broadly in their response to vaccination and subsequent challenge infection, with poor vaccine responders causing persistence of both		
3	infection and transmission in populations.		
4			
5	Keywords: Helminth; vaccine; immunoparasitology; machine learning; pathway ana-		
6	lysis; ageing.		

OUTLINE

Two main classes of vaccination: [AD, DA] vs [DD]. DD is the booster (can we rescue poor responders?). We are focussing on the variation within vaccine regimens: why do individuals vary despite identical vaccines?

Aims:

- what drives variation in immune response?
- why do vaccine lose efficacy in the wild?
- can we recover the non-responders, with boosters, diet, other interventions...
- vaccination to probe wild immune systems
- methods development

1. Partition differences between adjuvant and dose regimen

- Be clear that we chose alum to elicit type 2 antibody responses, because we can really only measure IgG1 in this system. Discussion: we might see different relationships between diet and antibody response if we measure other antibody isotypes.
- Comparison between AD, DA, DD

2. Total effects of interventions

- $OD \sim V + W + D + V : W + V : D + W : D$ on all mice => plot of OD by Diet (D) and Lab vs Wild (W)
- $OD \sim V + P + D + V : P + V : D + P : D$ on wild mice only
- Depending on the results of the models above, break down further.

- 27 3. Disentangling effects of environmental covariates with SCM: how do Sex, Weight,
28 Parasite infection, reproductive status, and time since vaccination affect immunisation
29 efficacy?

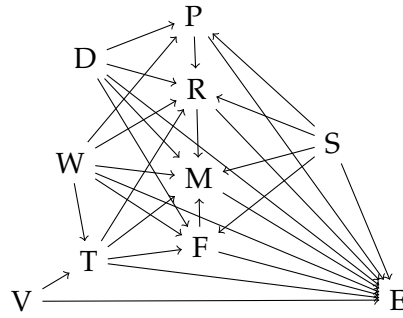


Fig. 1 Structural Causal Model

INTRODUCTION

Individuals vary widely in their responses to vaccination and little is known about the causes underlying low vaccine efficacy, nor how effective a vaccine will be “in the real world” against the pathogen for which it has been developed.

RESULTS

METHODS

Data processing

DT-specific antibody ELISA optical densities E were $\log_{10}$ -transformed, and along body weight M and days post-vaccination T , were centred around the mean and standardised to unit variance. Fat scores F are the sum of pelvic and dorsal five-point scales, and were modelled using a Dirichlet distribution. All other variables are treated as binary.

Structural Causal Models

Model construction

Model structure validation

To validate the causal structure hypothesised above, we tested the independencies it implies following chain, fork and collider rules^{1,2}. Specifically, we used the observed values for each of the nodes of the causal graph (Fig. 1) to test the truth of the following 15 implied independences derived from the graph: $(D \perp\!\!\!\perp S)$, $(D \perp\!\!\!\perp T)$, $(D \perp\!\!\!\perp V)$, $(D \perp\!\!\!\perp W)$, $(F \perp\!\!\!\perp P \mid D, R, S, T, W)$, $(F \perp\!\!\!\perp V \mid T, W)$,

($M \perp\!\!\!\perp P \mid D, R, S, T, W$), ($M \perp\!\!\!\perp V \mid T, W$), ($P \perp\!\!\!\perp T \mid W$), ($P \perp\!\!\!\perp V$), ($R \perp\!\!\!\perp V \mid T, W$), ($S \perp\!\!\!\perp T$), ($S \perp\!\!\!\perp V$), ($S \perp\!\!\!\perp W$), and ($V \perp\!\!\!\perp W$). For each of the tests above, we constructed a generalised linear model with the appropriate link function and considered the statement true if $P_X > 0.1$, where X is the response variable of interest (Supplementary Data: SCM construction).

Model parameterisation

Choice of Bayesian priors. Priors were chosen by simulating draws from priors and comparing them to observed data, with variance sufficiently large to contain all biologically plausible values for each variable.

Missing data. Twenty-two mice had no fat scores and one lacked a weight measurement. Their values were imputed from the priors updated by the full structural causal model, where $F := f(T, W, D, S)$ and $M := f(T, W, D, R, S)$ (Fig. 1).

Model updating and sampling.

ACKNOWLEDGEMENTS

COMPETING INTERESTS

The authors declare that there are no competing interests.

AUTHOR CONTRIBUTIONS

REFERENCES

1. Pearl, J. *Causality* (Cambridge University Press, 2009).
2. Cinelli, C., Forney, A. & Pearl, J. A Crash Course in Good and Bad Controls (2021).

FIGURE LEGENDS

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SUPPLEMENTARY INFORMATION

Supplementary Data: SCM construction

`1_Bayesian_SCM_construction.ipynb`

Supplementary Data: SCM specification

`2_Bayesian_SCM_specification.ipynb`

Supplementary Data: Wood mouse data

Supplementary Figures